

Julius M. Pfadt

Postdoctoral Researcher (University of Amsterdam) — Co-founder JASP Services BV

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About

I am a researcher in the lab of Eric-Jan Wagenmakers at the University of Amsterdam and a former DFG Walter-Benjamin fellow. My research centers on (Bayesian) statistical modeling, in particular, psychometrics, reliability estimation, and structural equation modeling. I develop open-source tools for JASP (BFpack, Factor, Reliability, SEM) and the R package Bayesrel, and I co-founded JASP Services BV. I am committed to open science and accessible methodology.

Education

- 2018–2023 **Doctorate (summa cum laude)**, Department of Psychological Research Methods, Ulm University
Project title: *The Present and Future of Reliability Analysis: Advances in Theory and Practice* (Link)
Supervisor: Prof. Dr. Morten Moshagen
- 2016–2018 **Master of Science**, Psychology, Ulm University, Germany
- 2012–2016 **Bachelor of Science**, Psychology, Ulm University, Germany

Experience

- 2025–present **Co-Founder of JASP Services BV**
- 2024–present **Postdoctoral Researcher**, Programme Group Psychological Methods, University of Amsterdam, The Netherlands
- 2023 **Postdoctoral Researcher**, Department of Methodology and Statistics, Tilburg University, The Netherlands
- 2019–present **Developer and Maintainer**, JASP, University of Amsterdam, The Netherlands
Design, implementation, and maintenance of statistical modules
- 2017 **Research Intern**, Programme Group Psychological Methods, University of Amsterdam, The Netherlands
Topic: Bayesian statistics
Supervisor: Prof. Dr. Eric-Jan Wagenmakers

Publications

- 2026 **Pfadt, J. M.**, Bartoš, F., Godmann, H. R., Waaijers, M., Groot, L., Heo, I., Mensink, L., Nak, J., De Ruiter, J. P., Sarafoglou, A., Siepe, B. S., Arena, G., Akrong, E., Aust, F., van den Bergh, D., Brenner, W., Doekemeijer, R. A., Donzallaz, M. C., van Doorn, J., Echevarria, N. O., . . . Wagenmakers, E.-J. (2026). *A methodological metamorphosis: The rapid rise of Bayesian inference and open science practices in psychology*. PsyArXiv. https://doi.org/10.31234/osf.io/ck3js_v2

- Mulder, J., & **Pfadt, J. M.** (2026). *Going in the right direction: A tutorial to directional hypothesis testing using the BFpack module in JASP*. PsyArXiv. https://doi.org/10.31234/osf.io/f582h_v3
- Pfadt, J. M.**, Merkle, E. C., & Wagenmakers, E.-J. (2026). *Bayes factors for structural equation models with bridge sampling and blavaan*. PsyArXiv. https://doi.org/10.31234/osf.io/pt2bc_v1
- Godmann, H. R., Molenaar, D., Ziegler, M., & **Pfadt, J. M.** (2026). *A tutorial on assessing measurement invariance with moderated (non-)linear factor analysis in JASP*. PsyArXiv. https://doi.org/10.31234/osf.io/6ftqg_v2
- Pfadt, J. M.**, Molenaar, D., Hurks, P., & Sijtsma, K. (2026). A tutorial on estimating the precision of individual test scores for anyone constructing and using psychological tests. *Psychometrika*, 1–22. <https://doi.org/10.1017/psy.2026.10081>
- 2025 Mulder, J., **Pfadt, J. M.**, & Wagenmakers, E.-J. (2025). A tutorial on Bayesian hypothesis testing of correlation coefficients using the BFpack-module in JASP. *Behavior Research Methods*, 57 (11), 311. <https://doi.org/10.3758/s13428-025-02846-5>
- 2023 **Pfadt, J. M.**, van den Bergh, D., & Moshagen, M. (2023). Classical and Bayesian uncertainty intervals for the reliability of multidimensional scales. *Structural Equation Modeling: A Multidisciplinary Journal*, 30 (3), 349–363. <https://doi.org/10.1080/10705511.2022.2124162>
- Pfadt, J. M.** (2023). *The present and future of reliability analysis: Advances in theory and practice* (Doctoral dissertation, Ulm University). <https://doi.org/10.18725/OPARU-49700>
- Sijtsma, K., & **Pfadt, J. M.** (2023). Reliability. In R. Tierney, F. Rizvi, & K. Ercikan (Eds.), *International encyclopedia of education* (4th ed., pp. 21–34). Elsevier. <https://doi.org/10.1016/B978-0-12-818630-5.10004-1>
- 2022 **Pfadt, J. M.**, van den Bergh, D., Sijtsma, K., Moshagen, M., & Wagenmakers, E.-J. (2022). Bayesian estimation of single-test reliability coefficients. *Multivariate Behavioral Research*, 57 (4), 620–641. <https://doi.org/10.1080/00273171.2021.1891855>
- Pfadt, J. M.**, van den Bergh, D., Sijtsma, K., & Wagenmakers, E.-J. (2022). A tutorial on Bayesian single-test reliability analysis with JASP. *Behavior Research Methods*, 55 (3), 1069–1078. <https://doi.org/10.3758/s13428-021-01778-0>
- Pfadt, J. M.**, & Sijtsma, K. (2022). Statistical properties of lower bounds and factor analysis methods for reliability estimation. In *Quantitative Psychology* (pp. 51–63). Springer International Publishing. https://doi.org/10.1007/978-3-031-04572-1_5
- 2021 Sijtsma, K., & **Pfadt, J. M.** (2021). Rejoinder: The future of reliability. *Psychometrika*, 86 (4), 887–892. <https://doi.org/10.1007/s11336-021-09807-9>
- Sijtsma, K., & **Pfadt, J. M.** (2021). Part II: On the use, the misuse, and the very limited usefulness of Cronbach's alpha: Discussing lower bounds and correlated errors. *Psychometrika*, 86 (4), 843–860. <https://doi.org/10.1007/s11336-021-09789-8>

Software

- 2019–present **Bayesrel**, R-package
Creator and maintainer of the R-package for Bayesian reliability estimation
- 2020–present **Modules in JASP: BFpack, Factor, Reliability, SEM**, Statistics program
Maintainer

Teaching

- 2024, 2026 **Structural Equation Modelling**, Lecturer, Research master course, University of Amsterdam, The Netherlands
- 2016–2019, 2021 **Applications of Multivariate Statistics in R**, Instructor, Master course, Ulm University, Germany

Talks

- Conference **Association for Psychological Science (APS) 2026 Annual Convention**, “The Past, Present, and Future of Scientific Reform”, Barcelona, Spain
- Conference **International Meeting of the Psychometric Society 2021**, “Bayesian Multidimensional Reliability”, online
- Invited **Research Section of the Forensic Psychiatry**, “Bayesian statistics”, Guenzburg, Germany

Funding, Awards & Scholarships

- 2024–2026 **Walter-Benjamin Fellowship** from the German Research Foundation (DFG) for “Bayesian Model Averaging for Structural Equation Models”
- 2025 **Open-Science Award** from the Open Science Community Amsterdam for “Tracking Change in Psychology: A Meta Research Project Using Human Coders and AI”

Reviewing

Ad-hoc reviewer for: *Behavior Research Methods*, *BMC Medical Research Methodology*, *BMC Musculoskeletal Disorders*, *British Journal of Mathematical and Statistical Psychology*, *Educational and Psychological Measurement*, *European Journal of Psychological Assessment*, *German Journal of Exercise and Sport Research*, *Journal of Mathematical Psychology*, *Perspectives on Psychological Science*, *Psychological Methods*, *Psychometrika*, *Structural Equation Modeling: A Multidisciplinary Journal*